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ENERGY

HOW INDIA CAN CAPITALISE ON ITS THORIUM RESERVES

Leveraging thorium to produce nuclear energy helps address both domestic needs and the country's export ambitions

India's atomic energy programme has had some very distinctive achievements. It has enabled us to emerge as a nuclear weapon state (including nuclear submarine capability) while also delivering pressurised heavy water reactors (PHWRs) and associated nuclear fuel cycle technology that is commercially successful and is performing with global standards. India now operates 19 PHWRs of 220, 540 and 700 MWe (MegaWatt electric) unit sizes with 14 more 700 MWe units under construction in fleet mode. We are also close to commissioning the 500 MWe prototype fast breeder reactor and its fuel cycle as the next step in our three-stage strategy, aimed at ensuring long-term energy security for India based on our vast thorium reserves, which are the largest in the world. Thorium is also the preferred fuel for high-temperature nuclear reactors required for producing cheaper clean hydrogen to meet industry needs.

India has only modest quantities of uranium, the only naturally available resource for obtaining fission energy. This has necessitated our thrust towards converting thorium into uranium-233. The fissile material so generated can efficiently produce energy in situ or in a sustainable breeder cycle with thorium. In uranium reactors, use of thorium also enables advantages such as augmented accident tolerance and improved reactor parameters that augment safety and proliferation resistance. Further, in PHWRs, thorium can increase the fuel burn-up to the level of other water reactors reducing the spent fuel arising by a factor of seven. These features can considerably soften the barriers to large-scale deployment of nuclear energy in emerging economy countries where the need for additional clean energy is the strongest in the context of realising net zero globally. India can thus become the energy capital of the world leveraging her thorium resources much the same way as major oil and gas-producing countries are today.

Viksit Bharat's energy requirement: Can this ever happen? The answer to this question lies in India's resolve to become a Viksit Bharat despite the net zero challenge and determination to develop thorium technologies for the purpose. To be *viksit,* one should realise a human development index of around 0.95, on par with developed nations. That would require per capita energy access beyond a minimum threshold. This threshold is around 2,400 kgoe (kilograms of oil equivalent) per year per capita. Since one expects a significant movement towards better energy efficiency through greater use of electricity and other measures, this threshold could significantly come down. Assuming that it would come down to around 60 per cent, the total energy requirement of Viksit Bharat would be around 28,000 TWh/yr (terawatt hour per year).